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Unit 3 : Session 2 : SLO 2 — Ripple-Carry vs Carry Look-ahead Adders

■ Section 1: The Ripple Effect (Understand)

- When you push the first domino it falls and knocks over the next one, which knocks over the one after it, and so on; there is a delay as the fall propagates down the line.
- Similarly, when adding $1111 + 0001$, the first full adder generates a carry that must ripple to the second stage, which generates a carry for the third, and so on. The final correct sum is available only after the carry has propagated through all stages — just like waiting for the last domino to fall.

■ Section 2: Identifying Limitations (Analyze) — single full adder delay = 10 ns

- Total propagation delay for a 4-bit ripple-carry adder: $4 \times 10 \text{ ns} = 40 \text{ ns}$
- Total propagation delay for a 16-bit ripple-carry adder: $16 \times 10 \text{ ns} = 160 \text{ ns}$
- Major limitation: the delay grows linearly with the number of bits. A 64-bit adder would need about 640 ns, making the processor too slow for modern high-speed applications.

■ Section 3: Introducing a Faster Solution (Remember/Understand)

- A carry look-ahead adder improves performance by reducing the propagation (1) DELAY.
- Instead of waiting for the carry to (2) RIPPLE through each full adder, the CLA uses special logic to calculate all the carry bits (3) SIMULTANEOUSLY.
- It does this by looking at the input bits and determining if a carry will be (4) GENERATED (created at that stage) or (5) PROPAGATED (passed from a previous stage).
- This makes the calculation of each carry (6) INDEPENDENT of the previous carry bit, speeding up the entire addition process.

■ Section 4: Concept Matching (Analyze)

1. Ripple-Carry Adder → B. Carry-out of one stage is the carry-in of the next, creating a chain reaction
2. Carry Look-ahead Adder → C. Uses complex logic to calculate all carries in parallel, improving speed
3. Propagation Delay → A. Time for an input change to produce a stable output change
4. Carry 'Generate' (gi) → E. A carry is created at a bit stage — both input bits are 1
5. Carry 'Propagate' (pi) → D. A carry is passed through a bit stage — one input is 1 and the other is 0

■ Section 5: Justifying the Choice (Evaluate) — high-performance gaming CPU

- Chosen adder: Carry Look-ahead Adder.
- Why better: calculation speed is critical for a gaming CPU. A CLA computes all carry bits in parallel, so it is significantly faster than a ripple-carry adder; arithmetic completes quickly, preventing lag.
- Drawback of the ripple-carry adder: its speed falls as the number of bits grows (e.g., 64-bit calculations) because of the long carry-ripple propagation delay, which would bottleneck the CPU.